MIL-G-51145A

1 February 1967
SUPERSEDING
MIL-G-51145
27 February 1963

MILITARY SPECIFICATION

GLAZING COMPOUND, BAKING PAN

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification establishes requirements for a baking type of glazing compound.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-704 - Pails: Shipping, Steel (1 through 12-Gallon).
PPP-D-729 - Drums: Metal, 55-Gallon (For Shipment of Non-Corrosive Material).

STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

MILITARY

MIL-STD-1005 - Renovation, Cleaning and Glazing of Bake Pans.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 6850

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(Copies of the specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Food and Drug Administration

Federal Food, Drug, and Cosmetic Act,

Food Additive Amendment, 21 Code

Federal Regulations 121.2510 For Contact with Food

(Application for copies should be addressed to Department of Health, Education, and Welfare, Food and Drug Administration, 330 Independence Ave., S. W., Washington 25, D.C.)

3. REQUIREMENTS

3.1 Material.-

3.1.1 Glazing compound.- The glazing compound shall be a clear non-pigmented baking type silicone resin coating of a grade designed and in commercial use, for the glazing of bread pans. The compound shall conform with all requirements of the Federal Food, Drug and Cosmetic Act, Food Additive Amendment, 21 Code Federal Regulations 121.2510, for Contact with Food, as applicable.

3.2 Performance and product characteristics.-

3.2.1 Cured coated glazing compound.- The cured coated glazing compound as used in MIL-STD-1005 shall not be marred or show haze or softening when tested as specified in 4.3.

3.2.2 Solids content.- The nonvolatile solids content shall be 20 $\pm$ 1 percent when tested as specified in 4.3.

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3.2.3 Viscosity.- The viscosity of the glazing compound shall be between 3 and 12 centipoises at 25° ± 1°C. when tested as specified in 4.3.

3.2.4 Specific gravity.- The specific gravity shall be 0.880 to 0.930 at 25° ± 1°C. when tested in accordance with 4.3.

3.2.5 Flash point.- The flash point shall be a minimum of 81°F. when tested in accordance with 4.3.

3.3 Workmanship.- The finished glazing compound shall be uniform and homogeneous conforming with the quality of product established by this specification, and the occurrence of defects shall not exceed the specified acceptable quality levels. Containers shall be free from dents or defects of manufacture and shall be clean, free of rust, corrosion and scale.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Inspection.- Inspection shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item.- Examination of the end item shall be made in accordance with the classification of defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.2.5. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers of glazing compound of the same type for examination under 4.2.2.1, 4.2.2.2, 4.2.2.3 and in units of shipping containers for examination under 4.2.2.4.

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4.2.2.1 Examination of the end item container for defects in closure, appearance, markings, construction and workmanship.-

Examine	Defect
Container	Not specified type, size or style of container. Any evidence of leakage; seepage from any part of the container. Any hole, tear, puncture, scuff, dent or any other defect in the body of the container, which may effect serviceability. Closure not as specified. Exterior coating (when applicable) not as specified.
Marking (including precautionary markings)	Printing or stenciling omitted, illegible, incorrect, incomplete, smeared.

4.2.2.2 Examination of the end item container for defects in appearance of glazing compound.- The sample unit for this examination shall be one filled unit container.

Examine	Defect
Appearance of glazing compound	Presence of dirt or foreign matter. Not homogeneous. Not clear. Contains lumps.

4.2.2.3 Examination of the end item container for defects in net contents.- The sample unit for this examination shall be one filled container. The average net contents shall be not less than the specified or indicated quantity on the container.

4.2.2.4 Examination of packing and markings.- An examination shall be made to determine that the packing and marking comply with the requirements of Section 5 of this specification. The sample unit for this examination shall be one shipping container fully packed selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. This examination does not apply when the primary container is the shipping container.

Examine	Defect
Packing	Not in accordance with contract requirements. Not level specified. Inadequate application of components, such as incomplete closures of containers.

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Examine	Defect
Packing (cont'd)	Any nonconforming component, component missing, damaged or otherwise defective, affecting serviceability.
Markings (including precautionary markings)	Omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.

* 4.2.2.5 Inspection levels and acceptable quality levels (AQLs) for examinations.- The inspection level for determining the sample size and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

	Examination	
Paragraph	Inspection level	AQL
4.2.2.1	S-3	2.5
4.2.2.2	S-3	1.5
4.2.2.3	S-2	---
4.2.2.4	S-2	2.5

* 4.2.3 Testing of the end item.- The end item shall be tested for characteristics listed in 4.3. The lot shall be expressed in gallons of glazing compound. The sample unit shall be one quart of glazing compound. The sample size shall be determined as shown below. The test sample shall consist of a two-quart composite made up of equal portions taken from each sample unit. Duplicate determinations shall be performed on the composite sample with no failure to meet the specified requirements.

Lot size (gallons)	Sample size (quarts)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The Federal Food, Drug and Cosmetic Act requirements specified in 3.1.1 shall be satisfied by a certificate of compliance from the supplier, certifying tha the glazing compound furnished conforms with all requirements of the Federal Food, Drug and Cosmetic Act, Food Additive Amendment, 21 Code Federal Regulations 121.2510 for Contact With Food, as applicable. The certificate of compliance shall be accompanied by actual test, inspection or other verifiable quality data.

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4.3 Tests.-

TABLE I.- Instructions for testing the end item

Characteristic	Specification reference: Requirement	Specification reference: Test method	Requirements applicable to: Composite sample	Requirements applicable to: Lot average	Number of determinations per unit composite sample	Results reported as: Pass or fail 2/	Results reported as: Numerically to nearest 1/
Cured coated glazing compound	3.2.1	4.3.1	X	-	2 on composite	X	
Solids content	3.2.2	4.3.2	X	-	2 on composite	-	percent
Viscosity	3.2.3	4.3.3	X	-	Average of 2 on composite	-	centipoise
Specific	3.2.4	Method 4183, Fed. Test Method Std. 141	X	-	2 on composite	-	0.001
Flash point	3.2.5	Method 4291, Fed. Test Method Std. 141	X	-	2 on composite	X	

1/ Test report shall include all values on which records are based.

2/ If failure is indicated, report either description of failure or numerical point of failure as applicable.

4.3.1 Cure test.- A bake pan or representative metal test panel coated with the cured glazing coating shall be placed on a beaker containing boiling water for ten minutes coated side down. Immediately after removal, the surface of the steamed area shall be scratched with a fingernail.

4.3.2 Solids content.- The solids content shall be determined by Method 4044 of Fed. Test Method Std. 141 except that the final time and temperature shall be 3 hours at 275° ± 2°F.

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4.3.3 Viscosity.- The viscosity of the glazing compound shall be determined as specified in Method 4287 of Fed. Test Method Std. 141. The test shall be conducted using a No. 1 spindle at a speed of 20 revolutions per minute.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.1.1 Level A.- Glazing compound shall be furnished in 5-gallon pails conforming to type II, class 4 of PPP-P-704 or in 55-gallon drums conforming to type I of PPP-D-729.

5.1.2 Level B.- Glazing compound shall be furnished in 5-gallon pails conforming to type II, class 2 of PPP-P-704 or in 55-gallon drums conforming to type II of PPP-D-729.

5.1.3 Level C.- Glazing compound shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The glazing compound covered by this specification is intended for use in glazing of bake pans for the baking of bread, buns, rolls and other low-sugar content baker products.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Capacity of unit container (see 5.2).
(c) Selection of applicable level of packing (see 5.2).

* 6.3 Marginal notations.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Preparing activity:

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Project No. 6850-0294

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-G-51145A - Glazing Compound, Baking Pan

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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